

Let $\mathbf{x}$ be from some probability distribution $P(\mathbf{x})$, that is

$$\mathbf{x} \sim P(\mathbf{x})$$

Also, let the target function be $f(\mathbf{x})$. Then we have

$$\begin{aligned} E_{\text{out}}(G) &= \mathbb{E}_{\mathbf{x} \sim P} [\mathbb{I}[f(\mathbf{x}) \neq G(\mathbf{x})]] \\ &= \mathbb{P}[f(\mathbf{x}) \neq G(\mathbf{x})] \end{aligned}$$

Observe that for some $\mathbf{x}$, if there are more than $M + 1$ different t such that $g_t(\mathbf{x}) = 1$, then

$$\begin{aligned} G(\mathbf{x}) &= \text{sign} \left(\sum_{t=1}^{2M+1} g_t(\mathbf{x}) \right) \\ &= \text{sign} \left(\overbrace{1 + 1 + \dots + 1}^{\geq 2M+1} \overbrace{-1 - 1 - \dots - 1}^{\leq 2M} \right) = 1 \end{aligned}$$

and $G(\mathbf{x}) = 1$

Similarly, for some $\mathbf{x}$, if there are more than $M + 1$ different t such that $g_t(\mathbf{x}) = -1$, then $G(\mathbf{x}) = -1$.

From this observation, we can conclude that for some $(y_n, \mathbf{x}_n) \in \mathcal{D}$, G wrongly classifies it if and only if there are more than $M + 1$ different t such that g_t wrongly classifies $(y_n, \mathbf{x}_n)$.

Hence

$$f(\mathbf{x}) \neq G(\mathbf{x}_n) \iff \sum_{t=1}^{2M+1} \mathbb{I}[f(\mathbf{x}) \neq g_t(\mathbf{w})] \geq M + 1$$

Therefore

$$\begin{aligned}
E_{\text{out}}(G) &= \mathbb{P}[f(\mathbf{x}) \neq G(\mathbf{x})] \\
&= \mathbb{P}\left[\sum_{t=1}^{2M+1} \mathbb{I}[f(\mathbf{x}) \neq g_t(\mathbf{x})] \geq M+1\right] \\
&= \sum_{m=M+1}^{2M+1} \left(\mathbb{P}\left[\sum_{t=1}^{2M+1} \mathbb{I}[f(\mathbf{x}) \neq g_t(\mathbf{x})] = m\right] \right) \\
&\leq \frac{1}{M+1} \sum_{m=M+1}^{2M+1} \left(m \cdot \mathbb{P}\left[\sum_{t=1}^{2M+1} \mathbb{I}[f(\mathbf{x}) \neq g_t(\mathbf{x})] = m\right] \right) \\
&\leq \frac{1}{M+1} \sum_{m=1}^{2M+1} \left(m \cdot \mathbb{P}\left[\sum_{t=1}^{2M+1} \mathbb{I}[f(\mathbf{x}) \neq g_t(\mathbf{x})] = m\right] \right) \\
&= \frac{1}{M+1} \mathbb{E}_{x \sim P} \left[\sum_{t=1}^{2M+1} \mathbb{I}[f(\mathbf{x}) \neq g_t(\mathbf{x})] \right] \\
&= \frac{1}{M+1} \sum_{t=1}^{2M+1} \mathbb{E}_{x \sim P} [\mathbb{I}[f(\mathbf{x}) \neq g_t(\mathbf{x})]] \\
&= \frac{1}{M+1} \sum_{t=1}^{2M+1} E_{\text{out}}(g_t) \\
&= \frac{1}{M+1} \sum_{t=1}^{2M+1} e_t
\end{aligned}$$

So $\frac{1}{M+1} \sum_{t=1}^{2M+1} e_t$ is an upper bound for $E_{\text{out}}(G)$

We then prove that this bound is tight by constructing a case when $E_{\text{out}}(G)$ meets this bound.

Consider the case where $M = 1$, $\mathbf{x}$ is one-dimensional, and

$$\begin{aligned}
P(x) &= U[0, 1] \quad (U \text{ is the uniform distribution}) \\
f(x) &= 1 \\
g_1(x) &= \begin{cases} +1 & x \leq \frac{1}{3} \\ -1 & \text{else} \end{cases} \\
g_2(x) &= \begin{cases} +1 & \frac{1}{3} < x \leq \frac{2}{3} \\ -1 & \text{else} \end{cases} \\
g_3(x) &= \begin{cases} +1 & \frac{2}{3} < x \\ -1 & \text{else} \end{cases}
\end{aligned}$$

In this case, we can see that $e_1 = e_2 = e_3 = \frac{2}{3}$, and $E_{\text{out}}(G) = 1$ (at least two of g_1, g_2, g_3 are wrong).

So

$$\frac{1}{M+1} \sum_{t=1}^{2M+1} e_t = \frac{\frac{2}{3} \cdot 3}{2} = 1 = E_{\text{out}}(G)$$

and the bound is met.

Therefore, we have proven that this is the tightest bound.

Q.E.D.

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We know that in iteration t , $\mathbf{u}$ is updated in the following manner

$$u_n^{(t+1)} \leftarrow \begin{cases} u_n^{(t)} \cdot \sqrt{\frac{1-\epsilon_t}{\epsilon_t}} & \llbracket y_n \neq g_t(\mathbf{x}_n) \rrbracket \\ u_n^{(t)} \cdot \sqrt{\frac{\epsilon_t}{1-\epsilon_t}} & \llbracket y_n = g_t(\mathbf{x}_n) \rrbracket \end{cases}$$

Therefore

$$U_{t+1} = \sum_{n=1}^N u_n^{(t+1)} \tag{1}$$

$$= \sqrt{\frac{1-\epsilon_t}{\epsilon_t}} \left(\sum_{n=1}^N \llbracket y_n \neq g_t(\mathbf{x}_n) \rrbracket u_n^{(t+1)} \right) + \sqrt{\frac{\epsilon_t}{1-\epsilon_t}} \left(\sum_{n=1}^N \llbracket y_n = g_t(\mathbf{x}_n) \rrbracket u_n^{(t+1)} \right) \tag{2}$$

$$= \sqrt{\frac{1-\epsilon_t}{\epsilon_t}} \left(\sum_{n=1}^N \llbracket y_n \neq g_t(\mathbf{x}_n) \rrbracket u_n^{(t+1)} \right) + \sqrt{\frac{\epsilon_t}{1-\epsilon_t}} \left(\sum_{n=1}^N (1 - \llbracket y_n \neq g_t(\mathbf{x}_n) \rrbracket) u_n^{(t+1)} \right) \tag{3}$$

Now consider the definition of ϵ_t

$$\begin{aligned} \epsilon_t &= \frac{\sum_{n=1}^N \llbracket y_n \neq g_t(\mathbf{x}_n) \rrbracket u_n^{(t)}}{\sum_{n=1}^N u_n^{(t)}} \\ \Rightarrow \sum_{n=1}^N \llbracket y_n \neq g_t(\mathbf{x}_n) \rrbracket u_n^{(t)} &= \epsilon_t \sum_{n=1}^N u_n^{(t)} \end{aligned}$$

And

$$\begin{aligned} 1 - \epsilon_t &= 1 - \frac{\sum_{n=1}^N \llbracket y_n \neq g_t(\mathbf{x}_n) \rrbracket u_n^{(t)}}{\sum_{n=1}^N u_n^{(t)}} \\ &= \frac{\sum_{n=1}^N (1 - \llbracket y_n \neq g_t(\mathbf{x}_n) \rrbracket) u_n^{(t+1)}}{\sum_{n=1}^N u_n^{(t)}} \\ \Rightarrow \sum_{n=1}^N (1 - \llbracket y_n \neq g_t(\mathbf{x}_n) \rrbracket) u_n^{(t+1)} &= (1 - \epsilon_t) \sum_{n=1}^N u_n^{(t)} \end{aligned}$$

Substituting into (3) we obtain

$$\begin{aligned} U_{t+1} &= \sqrt{\frac{1-\epsilon_t}{\epsilon_t}} \epsilon_t \sum_{n=1}^N u_n^{(t)} + \sqrt{\frac{\epsilon_t}{1-\epsilon_t}} (1 - \epsilon_t) \sum_{n=1}^N u_n^{(t)} \\ &= (2\sqrt{\epsilon_t(1-\epsilon_t)}) \sum_{n=1}^N u_n^{(t)} \\ &= 2\sqrt{\epsilon_t(1-\epsilon_t)} U_t \end{aligned}$$

Therefore

$$\frac{U_{t+1}}{U_t} = 2\sqrt{\epsilon_t(1 - \epsilon_t)}$$

Q.E.D.

The first part of the gradient boost decision tree is to find the hypothesis

$$\arg \min_h \frac{\eta}{N} \sum_{n=1}^N (h(\mathbf{x}_n) - (y_n - s_n))^2$$

Neglecting the constant terms, and utilizing the fact that $s_1 = s_2 = \dots = s_N = 0$ at the beginning of the first iteration, the problem becomes

$$\arg \min_h \sum_{n=1}^N (h(\mathbf{x}_n) - y_n)^2$$

Generally, the optimal hypothesis h that minimizes the error function above can be found with any regression algorithm, in this problem, our algorithm of choice is linear regression.

Hence, we have

$$h(\mathbf{x}_n) = \mathbf{w}^T \mathbf{x}'$$

Where

$$\begin{aligned} \mathbf{w} &= [w_0 \ w_1 \ \dots \ w_N]^T \\ \mathbf{x}' &= [1 \ \mathbf{w}]^T \end{aligned}$$

Therefore, the minimization problem becomes

$$\min_{\mathbf{w}} \sum_{n=1}^N (\mathbf{w}^T \mathbf{x}'_n - y_n)^2$$

Let $g_1(\mathbf{x})$ be the optimal hypothesis obtained by linear regression, at optimal, we have

$$\begin{aligned} \frac{\partial}{\partial w_i} \left(\sum_{n=1}^N (\mathbf{w}^T \mathbf{x}'_n - y_n)^2 \right) &= 0 \quad i = 0, 1, \dots, d \\ \Rightarrow \sum_{n=1}^N (g_1(\mathbf{x}_n) - y_n) x_{n_i} &= 0 \quad i = 0, 1, \dots, d \quad (\text{define } x_{n_0} = 1) \end{aligned}$$

Next, we proceed on the next step of the gradient boost decision tree, finding α_1

$$\begin{aligned}\alpha_1 &= \arg \min_{\eta} \frac{1}{N} \sum_{n=1}^N ((y_n - s_n) - \eta g_1(\mathbf{x}_n))^2 \\ &= \arg \min_{\eta} \sum_{n=1}^N (y_n - \eta g_1(\mathbf{x}_n))^2\end{aligned}$$

At optimal we have

$$\frac{\partial}{\partial \eta} \sum_{n=1}^N (y_n - \eta g_1(\mathbf{x}_n))^2 = \sum_{n=1}^N -g_1(\mathbf{x}_n)(y_n - \eta g_1(\mathbf{x}_n)) = 0$$

Hence

$$\begin{aligned}& \sum_{n=1}^N g_1(\mathbf{x}_n)(y_n - \alpha_1 g_1(\mathbf{x}_n)) = 0 \\ \Rightarrow & \alpha_1 \sum_{n=1}^N g_1(\mathbf{x}_n)^2 - \sum_{n=1}^N g_1(\mathbf{x}_n)y_n = 0 \\ \Rightarrow & \alpha_1 = \frac{\sum_{n=1}^N g_1(\mathbf{x}_n)^2}{\sum_{n=1}^N g_1(\mathbf{x}_n)y_n}\end{aligned}$$

Recall the result obtained earlier

$$\begin{cases} \sum_{n=1}^N (g_1(\mathbf{x}_n) - y_n)x_{n_0} = 0 \\ \sum_{n=1}^N (g_1(\mathbf{x}_n) - y_n)x_{n_1} = 0 \\ \vdots \\ \sum_{n=1}^N (g_1(\mathbf{x}_n) - y_n)x_{n_d} = 0 \end{cases}$$

Which leads to

$$\begin{cases} \sum_{n=1}^N g_1(\mathbf{x}_n)x_{n_0} = \sum_{n=1}^N y_n x_{n_0} \\ \sum_{n=1}^N g_1(\mathbf{x}_n)x_{n_1} = \sum_{n=1}^N y_n x_{n_1} \\ \vdots \\ \sum_{n=1}^N g_1(\mathbf{x}_n)x_{n_d} = \sum_{n=1}^N y_n x_{n_d} \end{cases}$$

By multiplying both sides of the i -th equation with w_i , where $\mathbf{w}$ is the weight vector of g_1 , and summing all the resulting equations, we can get

$$\begin{aligned}
& \sum_{i=0}^d \left(w_i \sum_{n=1}^N g_1(\mathbf{x}_n) x_{ni} \right) = \sum_{i=0}^d \left(w_i \sum_{n=1}^N y_n x_{ni} \right) \\
\Rightarrow & \sum_{n=1}^N \left(g_1(\mathbf{x}_n) \sum_{i=0}^d w_i x_{ni} \right) = \sum_{n=1}^N \left(y_n \sum_{i=0}^d w_i x_{ni} \right) \\
\Rightarrow & \sum_{n=1}^N g_1(\mathbf{x}_n) \cdot g_1(\mathbf{x}_n) = \sum_{n=1}^N y_n g_1(\mathbf{x}_n) \\
\Rightarrow & \sum_{n=1}^N g_1(\mathbf{x}_n)^2 = \sum_{n=1}^N y_n g_1(\mathbf{x}_n)
\end{aligned}$$

Lastly, it was previously derived that

$$\alpha_1 = \frac{\sum_{n=1}^N g_1(\mathbf{x}_n)^2}{\sum_{n=1}^N g_1(\mathbf{x}_n) y_n}$$

Since

$$\sum_{n=1}^N g_1(\mathbf{x}_n)^2 = \sum_{n=1}^N y_n g_1(\mathbf{x}_n)$$

we have proven

$$\alpha_1 = 1$$

Q.E.D.

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To make the derivation more comprehensive, we use $s_{n,t}$ and $s_{n,t+1}$ to denote s_n before and after being updated in iteration t , respectively.

In iteration t , after obtaining the hypothesis $g_t = h$ using some regression algorithm on $\{\mathbf{x}_n, y_n - s_{n,t}\}$, α_t is found by

$$\begin{aligned}\alpha_t &= \arg \min_{\eta} \frac{1}{N} \sum_{n=1}^N ((y_n - s_{n,t}) - \eta g_t(\mathbf{x}_n))^2 \\ &= \arg \min_{\eta} \sum_{n=1}^N ((y_n - s_{n,t}) - \eta g_t(\mathbf{x}_n))^2\end{aligned}$$

The optimal η can be found by solving

$$\frac{\partial}{\partial \eta} \left(\sum_{n=1}^N ((y_n - s_{n,t}) - \eta g_t(\mathbf{x}_n))^2 \right) = -2 \sum_{n=1}^N g_t(\mathbf{x}_n) (y_n - s_{n,t} - \eta g_t(\mathbf{x}_n)) = 0$$

Hence we have

$$\sum_{n=1}^N g_t(\mathbf{x}_n) (y_n - s_{n,t} - \alpha_t g_t(\mathbf{x}_n)) = 0 \quad (4)$$

After α_t and g_t is found, $s_{n,t}$ are updated into

$$s_{n,t} \rightarrow s_{n,t+1} = s_{n,t} + \alpha_t g_t(\mathbf{x}_n)$$

Therefore, after the update, we have

$$\begin{aligned}\sum_{n=1}^N (y_n - s_{n,t+1}) g_t(\mathbf{x}_n) &= \sum_{n=1}^N (y_n - (s_{n,t} + \alpha_t g_t(\mathbf{x}_n))) g_t(\mathbf{x}_n) \\ &= \sum_{n=1}^N (y_n - s_{n,t} - \alpha_t g_t(\mathbf{x}_n)) g_t(\mathbf{x}_n) \\ &= 0 \quad (\text{by (4)})\end{aligned}$$

Hence we have proven that $\sum_{n=1}^N (y_n - s_n) g_t(\mathbf{x}_n) = 0$ after updating all s_n in iteration t using the steepest η as α_t .

Q.E.D.

9

The neural network with one hidden layer and $\tanh$ as the transformation function at each layer has the output of the form

$$h(\mathbf{x}) = \tanh(s_1^{(2)})$$

$$s_1^{(2)} = \sum_{k=1}^{d^{(1)}} w_{k,1}^{(2)} x_k^{(1)}$$

and

$$x_i^{(1)} = \tanh(s_i^{(1)})$$

$$s_i^{(1)} = \sum_{k=1}^{d^{(0)}} w_{k,i}^{(1)} x_k^{(0)}$$

With the error function

$$e_n = (y_n - h(\mathbf{x}_n))^2$$

We then try to prove the following statement by induction

$$w_{i,j}^{(1)} = w_{i,j+1}^{(1)} \quad \text{for all } i \text{ and } 1 \leq j < d^{(1)} \quad (5)$$

$$w_{i,1}^{(2)} = w_{i+1,1}^{(2)} \quad \text{for all } 1 \leq i < d^{(0)} \quad (6)$$

Suppose that after iteration t the statement holds, then in iteration $t + 1$, we have

$$\begin{aligned} \frac{\partial e_n}{\partial w_{i,1}^{(2)}} &= \left(\frac{\partial e_n}{\partial s_1^{(2)}} \right) \left(\frac{\partial s_1^{(2)}}{\partial w_{i,1}^{(2)}} \right) \\ &= -2(y_n - \tanh(s_1^{(2)})) \tanh'(s_1^{(2)}) x_i^{(1)} \end{aligned}$$

and

$$\begin{aligned} \frac{\partial e_n}{\partial w_{i,j}^{(1)}} &= \left(\frac{\partial e_n}{\partial s_j^{(1)}} \right) \left(\frac{\partial s_j^{(1)}}{\partial w_{i,j}^{(1)}} \right) \\ &= \left(\frac{\partial e_n}{\partial s_1^{(2)}} \right) \left(\frac{\partial s_1^{(2)}}{\partial s_j^{(1)}} \right) x_i^{(0)} \\ &= \left(-2(y_n - \tanh(s_1^{(2)})) \tanh'(s_1^{(2)}) \right) \left(w_{j,1}^{(2)} \tanh'(s_j^{(1)}) \right) x_i^{(0)} \end{aligned}$$

First we analyze $\frac{\partial e_n}{\partial w_{i,1}^{(2)}}$, we can see that the only term dependent of i in its expression is $x_i^{(1)}$.

We also have

$$\begin{aligned} x_i^{(1)} &= \tanh(s_i^{(1)}) \\ &= \tanh\left(\sum_{k=1}^{d^{(1)}} w_{k,i}^{(1)} x_k^{(0)}\right) \end{aligned}$$

With condition (5), we know that $w_{k,i}^{(1)}$ is the same for all $1 \leq i < d^{(1)}$, so $s_i^{(1)} = \sum_{k=1}^{d^{(1)}} w_{k,i}^{(1)} x_k^{(0)}$ is independent of i .

Therefore, $x_i^{(1)}$ is also independent of i and $\frac{\partial e_n}{\partial w_{i,1}^{(2)}}$ is independent of i .

In the $t + 1$ -th iteration of the mini-batch gradient descent, $w_{i,1}^{(2)}$ is updated proportional to the average of $\frac{\partial e_n}{\partial w_{i,1}^{(2)}}$ over some set of different n . Since we know that for any particular n , $\frac{\partial e_n}{\partial w_{i,1}^{(2)}}$ is the same for all i , after the update $w_{i,1}^{(2)}$ is still the same for all i .

Hence condition (6) still holds after the $t + 1$ -th iteration.

Next, we analyze $\frac{\partial e_n}{\partial w_{i,j}^{(1)}}$, the only term dependent of j in its expression is $w_{j,1}^{(2)} \tanh'(s_j^{(1)})$

By condition (6), we know that $w_{j,1}^{(2)}$ is the same for all $1 \leq j \leq d^{(1)}$

Also, in the previous part we have proven that $s_i^{(1)}$ is the same for all $1 \leq i \leq d^{(1)}$

Therefore, we can conclude that $\frac{\partial e_n}{\partial w_{i,j}^{(1)}}$ is independent of j .

Similarly, in the $t + 1$ -th iteration $w_{i,j}^{(1)}$ is updated proportional to the average of $\frac{\partial e_n}{\partial w_{i,j}^{(1)}}$ over some different n . Since $\frac{\partial e_n}{\partial w_{i,j}^{(1)}}$ is independent of j , $w_{i,j}^{(1)} = w_{i,j+1}^{(1)}$ for all i and $1 \leq i < d^{(1)}$, and condition (5) still holds after the $t + 1$ -th iteration.

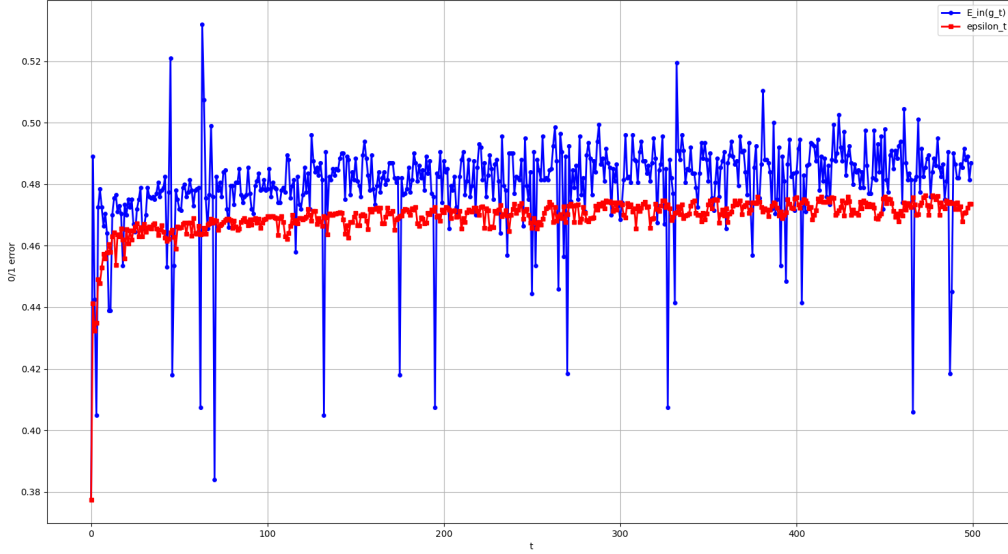
Lastly, at the beginning of the first iteration, all $w_{i,j}^l$ is set to the same value, 0.5, so the two conditions are satisfied.

Therefore by induction we've proven that $w_{i,j}^{(1)} = w_{i,j+1}^{(1)}$ for all i and $1 \leq j < d^{(1)}$ ((5))

Q.E.D.

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Plot:



Findings:

From the plot, we can see that the trend in $E_{\text{in}}(g_t)$ and in ϵ_t is similar. They both increase rapidly at the beginning of the iterations, stabilize at around $t = 10$, and remain at around the same level for the rest of the process.

We can see that the fluctuation in $E_{\text{in}}(g_t)$ is a lot larger than that in ϵ_t . A possible explanation to this phenomena is that since in every iteration, the best hypothesis is the one that results in the smallest ϵ_t , so the resulting minimal ϵ_t would be similar in value among iterations, as the minimizing ability of the decision given some example weight $\mathbf{u}$ should not differ much. On the other hand, the selection of the hypothesis is not as directly related to E_{in} , so although E_{in} has a similar trend, it is more prone to randomness, and its variation is a lot larger.

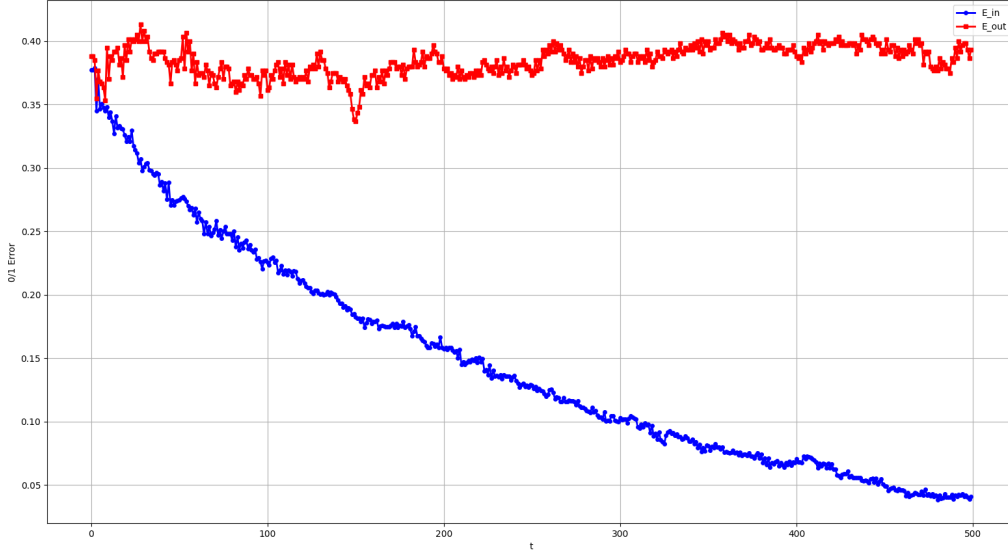
Also, in the lecture, we are taught that in every iteration, the weight for each example is multiplied or divided by $\sqrt{\frac{\epsilon_t}{1-\epsilon_t}}$, we can see that as ϵ_t gets closer to $\frac{1}{2}$, this value gets close to 1, which means that in each update the weights are not changed significantly. In other words, as ϵ_t approaches $\frac{1}{2}$, the algorithm converges to some point where more iterations no longer affect the output of the AdaBoost algorithm.

This analysis is consistent to what is shown in the graph, as we can see that after some iterations, ϵ_t gets closer to $\frac{1}{2}$, and the rate at which ϵ_t grow become much smaller. Also, the trend of E_{in} also becomes more stable, despite still having large deviations.

Code Snapshot:

11

Plot:



Findings:

First, we can see that $E_{in}(G_t)$ decreases steadily throughout the 500 iterations.

Starting at about 0.38 at $t = 0$, $E_{in}(G_t)$ drops to less than 0.05 at the end of the 500-th iteration. The significant decrease of $E_{in}(G_t)$ shows that although decision stump is a rather simple and weak model, adaptive boosting does indeed allow it to implement more complex boundaries to fit the training data set by aggregation.

Also, we can see that the rate at which $E_{in}(G_t)$ decreases is greater at small t than at large t . This is probably because as the number of iteration increases, the model approaches a limit beyond which aggregation cannot make the fitting performance better.

Second, we see that despite the significant decrease in $E_{in}(G_t)$, the decrease in $E_{out}(G_t)$ is really insignificant.

In most of the 500 steps, $E_{out}(G_t)$ fluctuates between 0.35 and 0.40. Although at about $t = 150$ $E_{out}(G_t)$ drops to a minima of approximately 0.34, $E_{out}(G_t)$ soon increases back to above 0.375 and remains in the range of 0.37 and 0.40 for the rest of the iterations.

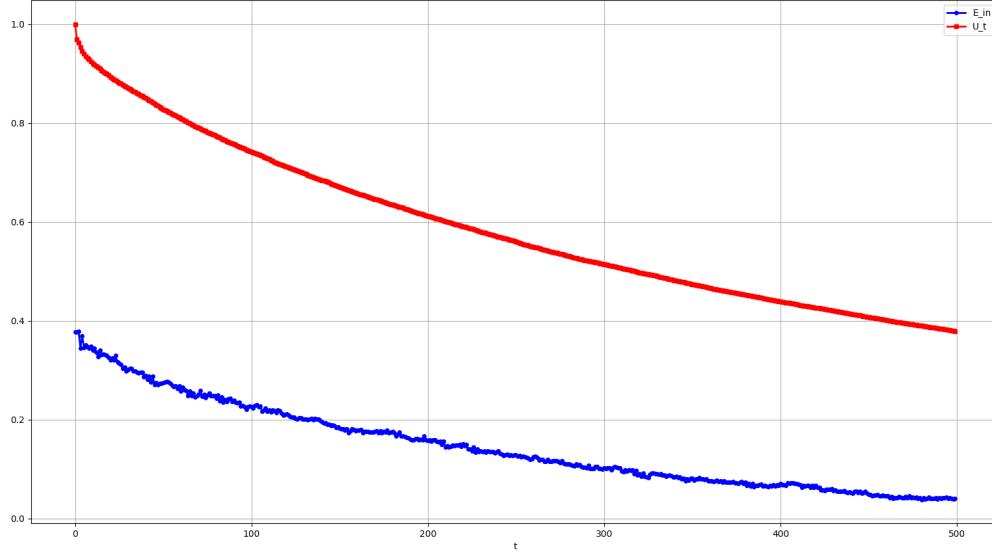
The insignificance, if not inexistence, of $E_{out}(G_t)$ shows that although AdaBoost combined with decision stump allows the model to implement complex boundaries and fit the training set, the improve in the performance on the testing set is greatly limited.

This phenomena might suggest that AdaBoost with decision stump might not be a great model regarding this data set.

Another thing worth noting is that at small t , the fluctuation of $E_{\text{out}}(G_t)$ is greater than that at large t . This might be because although not resulting in a better performance of classifying the testing data set, a larger number of iteration still leads to more stable performance of the model.

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Plot:



Findings:

Form the plot we can see that both U_t and E_{in} decreases steadily throughout the iterations. The reason for the decrease in E_{in} is described in P11.

For U_t , we have proven in problem 6 that $\frac{U_{t+1}}{U_t} = 2\epsilon_t\sqrt{\epsilon_t(1-\epsilon_t)}$, then, by AM-GM inequality, we have

$$\begin{aligned} \sqrt{\epsilon_t(1-\epsilon_t)} &\leq \frac{\epsilon_t + (1-\epsilon_t)}{2} \\ \Rightarrow 2\sqrt{\epsilon_t(1-\epsilon_t)} &\leq 1 \end{aligned}$$

Hence, U_t must decrease monotonically, which fits our observation from the plot.

Also, from the plot we can see that as t increases, both U_t and E_{in} decreases more slowly, and approaches some constant value asymptotically. This shows that the AdaBoost algorithm is indeed converging, and that after many number of iterations, the algorithm approaches a point where the parameters $\mathbf{u}$ stabilize, and E_{in} can be improved no more.